Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18MA5DE13]

FIFTH SEMESTER B.Tech. EXAMINATION, DECEMBER–2021

MOBILE DEVICE AND NETWORK ARCHITECTURE

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. In practical cellular radio systems, the MSC is programmed to allocate radio channels differently for the closest co-channel cells. This technique, called a hunting sequence, ensures that co-channel cells first use different channels from within the co-channel set, before the same channels are assigned to calls in nearby cells. This minimizes co-channel interference when the cellular system is not Kelly loaded. Consider 3 adjoining clusters, and design an algorithm that may be used by the MSC to hunt for appropriate channels when requested from co-channel cells. Assume a 7-cell reuse pattern with 3 sectors per cell, and use the U.S. cellular channel allocation scheme for the A-side carrier. **09**

b. Explain the expectations for a mobile application user and how this affects test prioritization. **05**

2. a. Why the traditional IP cannot be used in a mobile network, what are the main differences between the traditional IP and the mobile IP? How does mobile IP support mobile hubs? **07**

b. Draw a neat diagram and describe the complete hardware architecture of mobile handset. Is there any evaluation in the architecture of the Mobile handset? If yes, then justify your answer with the help of the comparison table. **07**

3. a. Describe the history of mobiles and mobile applications? Explain the Mobile Application Evolution in details **07**

b. Describe about the designing the right user interface for mobile application. **07**

OR

4. a. Being an Agile Developer, you should have some responsibilities regarding your team and product. You have to keep some major issues in your mind for the excellence. So, **10**

i. What should you look for in a Development Team? (2)
ii. Which is Better, Manual, Or Automated Testing? (2)
iii. How Do You Plan to Maintain Your App Post-Launch? (2)
iv. Why Is a Product Roadmap Important for Mobile App Development? (2)
v. How Much Does It Cost to Develop an App? (2)

	b.	What Is the Difference between Mobile Device Testing and Mobile Application Testing?	**04**
5.	**a.**	How will you convert text into voice in both Android and iOS? Explain with examples.	**06**
	b.	For a Mobile App Developer, how many Mobile App Programming languages are available for both Android and iOS.	**08**
		i. Consider yourself as a Mobile App Developer (Android) which language is most suitable for designing.	
		ii. Consider yourself as a Mobile App Developer (iOS) which language is most suitable for designing.	

OR

6.	**a.**	How the infrastructure and resources are managed in mobile applications? Discuss it.	**07**
	b.	Every time you send an SMS message to the emulator (or device), your application and the built – in application both are receiving it but you want that only your application receives it. How you will handle this situation and prevent the built – in application to receive it?	**07**
7.	**a.**	Explain Web Service Architecture. What are the different layers of the web service protocol stack?	**06**
	b.	Can you make an app multilingual? What programming languages do you work with to develop native apps? Will your app work as have designed on iOS or Android?	**08**

OR

8.	**a.**	Which type of third party application uses AWS APIs to control the various components and instances you have running? Explain all the applications in details with real world examples.	**10**
	b.	Explain Web Application Architecture in detail.	**04**